

Magnetism & Earth's Field

Important Terms:

- ⇒ Magnetizing field: $\vec{H} = \frac{\vec{B}}{\mu}$ (A/m). $\vec{H}$ is independent of medium.
- ⇒ Intensity of magnetization: $I = \frac{\text{Magnetic Moment}}{\text{Volume}} \Rightarrow I = \frac{M}{V}$.
- ⇒ Magnetic susceptibility: $\chi = \frac{I}{H}$.
- ⇒ Magnetic permeability: The measure of the degree to which the lines of force can penetrate or permeate the medium.

↳ Fundamental relations:

$$B = \mu_0 H + \mu_0 I = \mu H$$

↳ Relative Permeability: $\mu_r = \frac{\mu}{\mu_0} = 1 + \chi$.

• Components of Earth's Magnetic Field:

⇒ θ = Angle of declination (Angle between G.M and M.M).

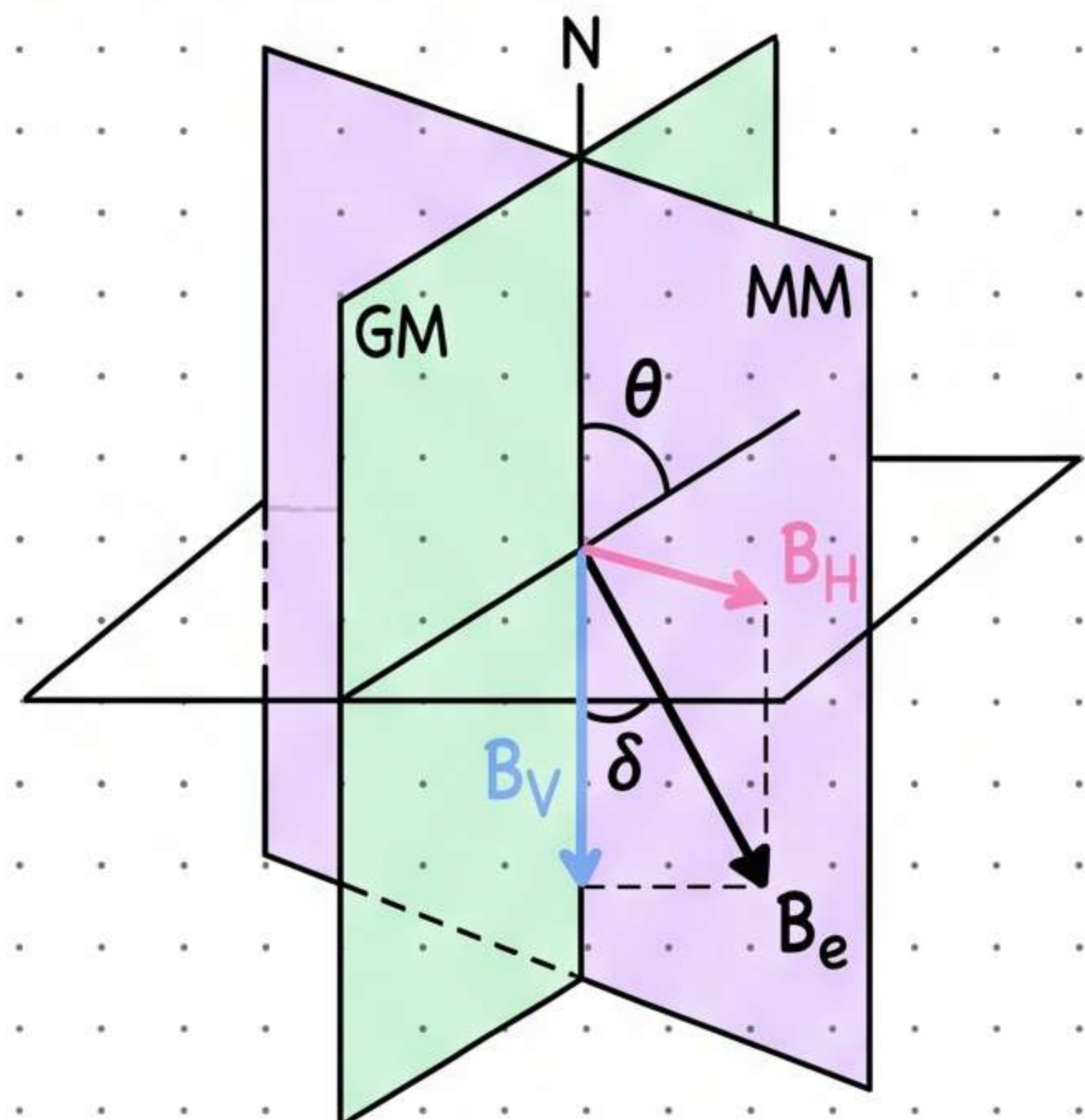
⇒ δ = Angle of dip.

⇒ Relations:

$$\tan \delta = \frac{B_V}{B_H}$$

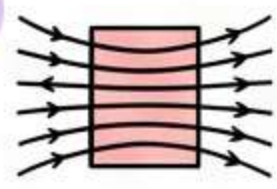
$$B_e = \frac{B_H}{\cos \delta}$$

$$B_e = \sqrt{B_V^2 + B_H^2}$$



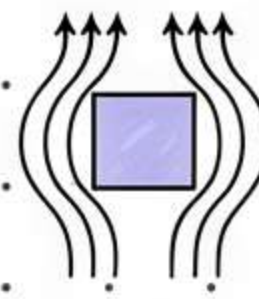
Magnetic Materials Classification

Paramagnetic Materials



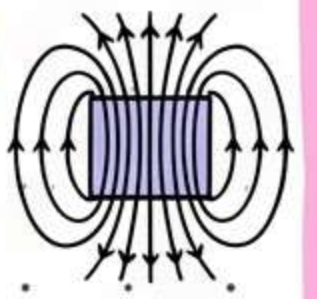
- Feeble Magnetisation along $\vec{H}$.
- Susceptibility: $0 < \chi < 1$.
- Relative Permeability: $\mu_r > 1$.
- Temperature Dependence: $\chi \propto \frac{1}{T}$.

Diamagnetic Materials



- Feeble Magnetisation opposite of $\vec{H}$.
- Susceptibility: $-1 \leq \chi < 0$.
- Relative Permeability: $0 < \mu_r < 1$.
- Temperature Dependence: χ is Independent of T .

Ferromagnetic Materials

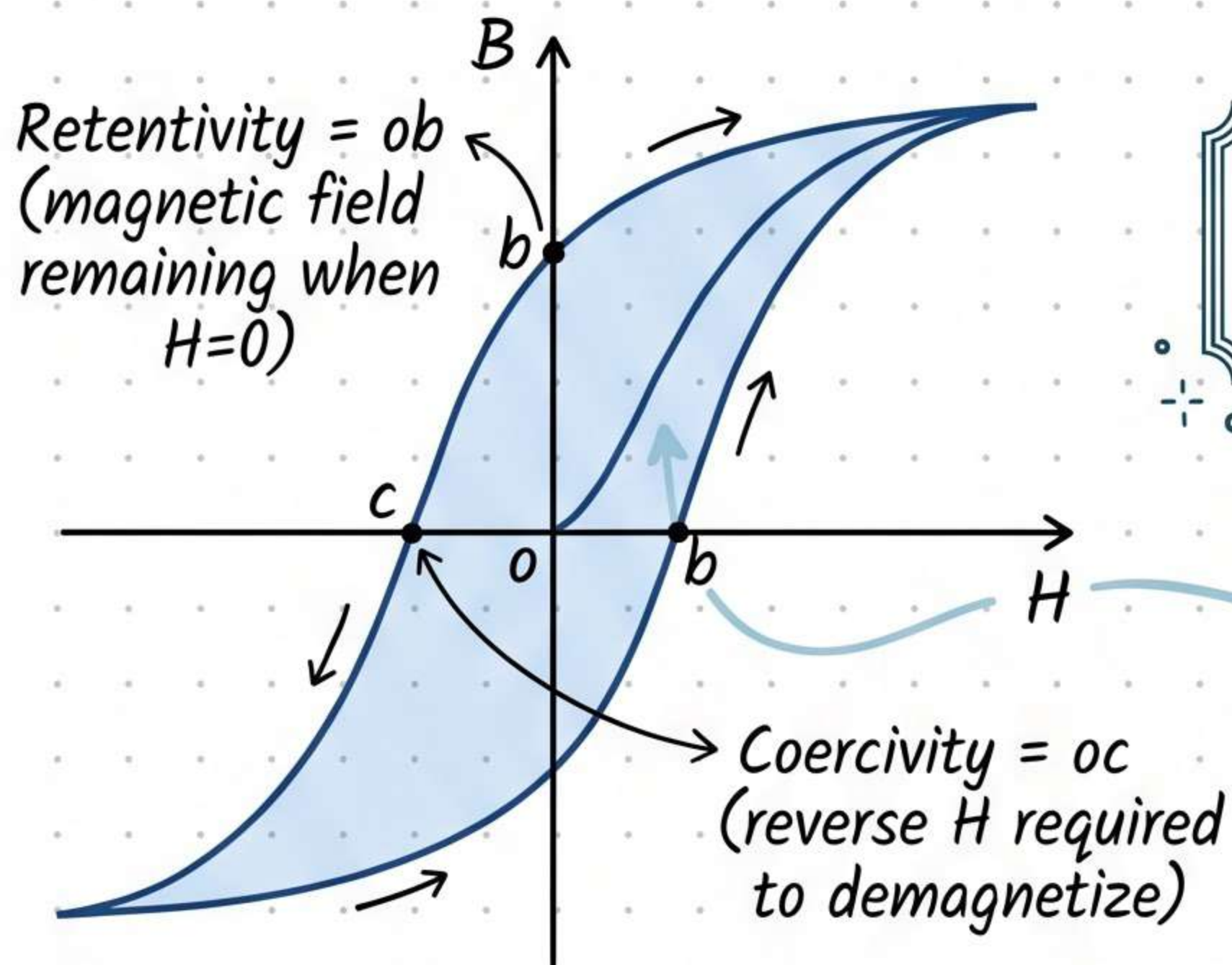


- Strong Magnetisation along $\vec{H}$.
- Susceptibility: χ is of the order 10^3 .
- Relative Permeability: $\mu_r \gg 1$.
- Temperature Dependence:

$$\chi = \frac{C}{T - T_c} \text{ for } T > T_c.$$

→ Here $T_c \rightarrow$ Curie temperature (above which a ferromagnetic material becomes paramagnetic).

Hysteresis & Bar Magnet Fields



Bar Magnet Fundamentals

$$\text{Magnetic Moment: } \vec{M} = m \cdot 2\ell$$

Area of loop = energy loss per unit volume per cycle

Field due to Bar Magnet: Axial Point (P)

$$B_P = \frac{\mu_0}{4\pi} \cdot \frac{(4md)\ell}{(d^2 - \ell^2)^2}$$

$$= \frac{\mu_0}{4\pi} \cdot \frac{2Md}{(d^2 - \ell^2)^2}$$

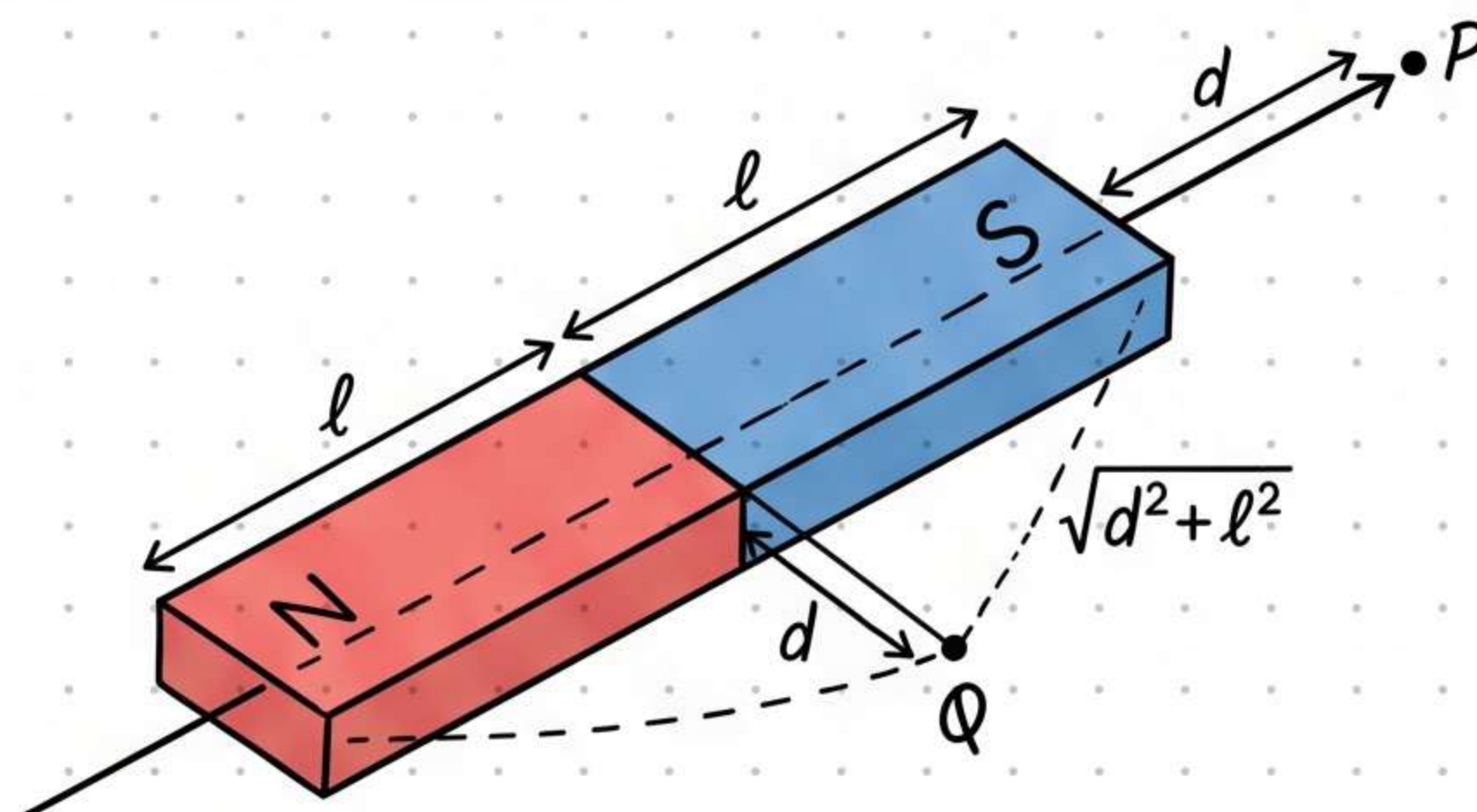
$$\text{If } \ell^2 \ll d^2 \Rightarrow B_P = \frac{\mu_0}{4\pi} \cdot \frac{2M}{d^3}$$

Field due to Bar Magnet: Equatorial Point (Q)

$$B_Q = \frac{\mu_0}{4\pi} \cdot \frac{2m\ell}{(d^2 + \ell^2)^{3/2}}$$

$$= \frac{\mu_0}{4\pi} \cdot \frac{M}{(d^2 + \ell^2)^{3/2}}$$

$$\text{If } \ell^2 \ll d^2 \Rightarrow B_Q = \frac{\mu_0}{4\pi} \cdot \frac{M}{d^3}$$



Magnets in External Fields

Field at a General Point (r, θ)

Radial component:

$$B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2M \cos \theta}{r^3} \quad \text{PDF}$$

Transverse component:

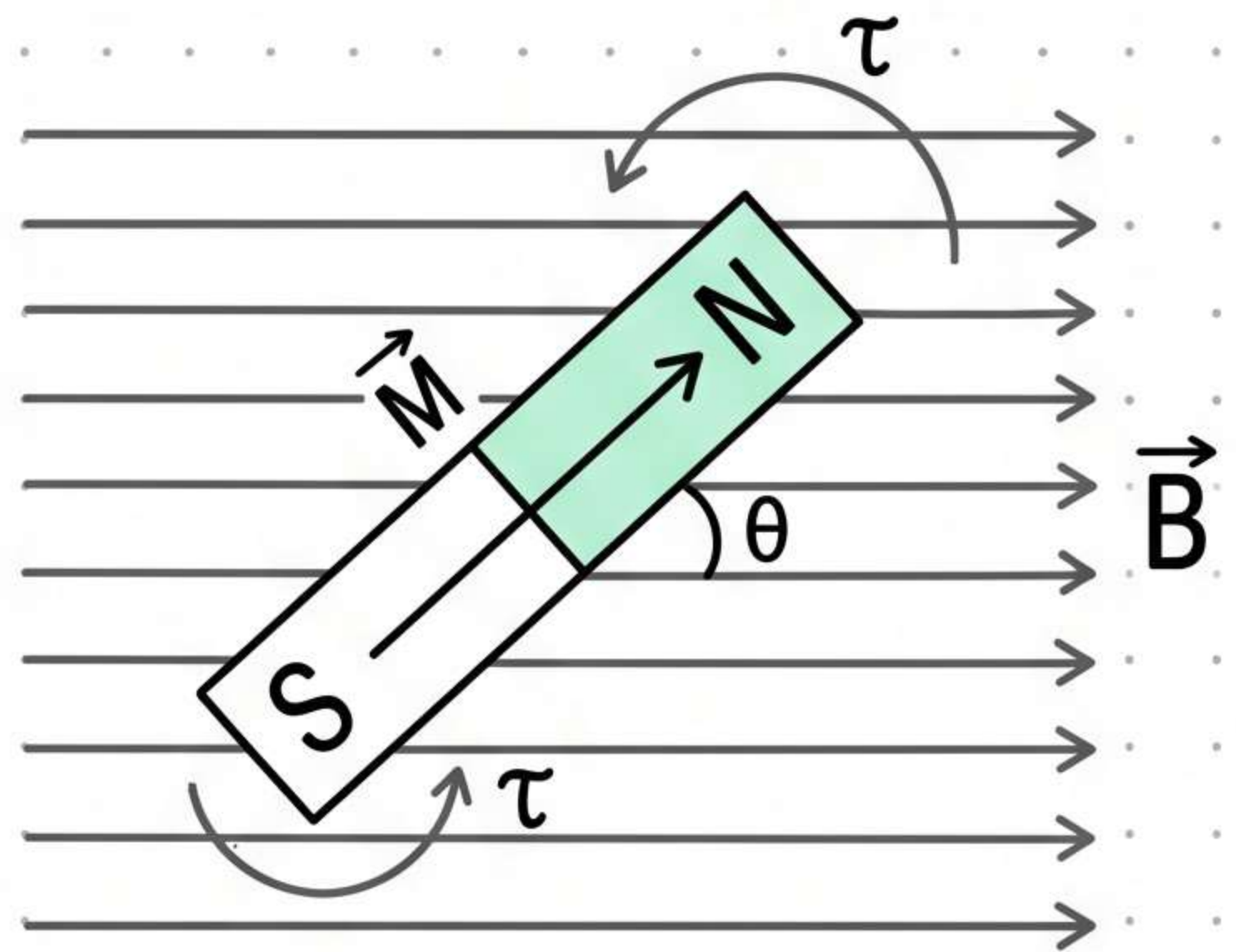
$$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{M \sin \theta}{r^3} \quad \text{PDF}$$

Net Magnetic Field:

$$B = \frac{\mu_0}{4\pi} \cdot \frac{M}{r^3} \cdot \sqrt{1 + 3 \cos^2 \theta} \quad \text{PDF}$$

Direction:

$$\tan \phi = \frac{\tan \theta}{2} \quad \text{PDF}$$



Bar Magnet in Uniform Magnetic Field

Torque on bar magnet: $\vec{\tau} = \vec{M} \times \vec{B}$ PDF

Potential energy: $U = -\vec{M} \cdot \vec{B}$ PDF

Work done in rotating the bar magnet: $W_{\text{ext}} = MB(\cos \theta_i - \cos \theta_f)$ PDF

Time period of small oscillations: $T = 2\pi \sqrt{\frac{I}{MB}}$ (where $I \rightarrow$ Moment of inertia of bar magnet) PDF